CoopInfer: Host–Device Cooperative Inference Scheduling

A strategy-driven prototype for DAG partitioning and schedule visualization

CoopInfer Project

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Motivation

- Embodied AI pipelines often run as DAGs of perception, fusion, and control operators.
- Some operators must stay on the robot-side device, while others may run on an edge host.
- The scheduling problem is not only graph partitioning:
  - cross-device edges incur network transfer cost;
  - each side has a finite execution queue;
  - end-to-end latency constraints can make otherwise low-loss solutions infeasible.
- CoopInfer is a desktop prototype for editing, solving, and visualizing these schedules.

System Overview

Inputs

- DAG nodes with device/host compute costs.
- Edges with transfer sizes.
- Bandwidth, latency, objective weight.
- Optional E2E latency limit.

Outputs

- Node placement: $x_i = 0$ for $\textsc{Device}$, $x_i = 1$ for $\textsc{Host}$.
- End-to-end latency and device utilization.
- SVG topology and profiler-style schedule timeline.

Data Model

- In memory, topology is stored in a `networkx.DiGraph`.
- Node attributes:
  - `id`, `name`, `c_dev`, `c_host`, `fixed_dev`, `x`.
- Edge attributes:
  - `source`, `target`, `size` in MB.
- Environment fields:
  - `bandwidth`, `latency`, `weight_latency`, `latency_limit`.
- JSON load/save keeps experiments reproducible and shareable.

Evaluator: Data Dependency + Resource Queues

For each node i in topological order:

$$DataReady_i = \max_{j \in pred(i)} \left(F_j + \mathbb{I}(x_j \neq x_i) \left(L + \frac{s_{ji}}{B} \right) \right)$$

The evaluator then waits for the selected worker queue:

$$S_i = \begin{cases} \max(DataReady_i, time\_dev\_ready), & x_i = 0 \\ \max(DataReady_i, time\_host\_ready), & x_i = 1 \end{cases}$$

$$F_i = S_i + (1 - x_i) c_i^{dev} + x_i c_i^{host}$$

The E2E latency is $\tau = \max_i F_i$.

Objective and Feasibility

The solver minimizes a weighted objective:

$$J = w \cdot \frac{\tau - \tau_{min}}{\tau_{max} - \tau_{min}} + (1 - w)\left(1 - \frac{\sum_i (1 - x_i) c_i^{dev}}{\sum_i c_i^{dev}}\right)$$

Latency limit

A positive `latency_limit` acts as a hard feasibility constraint:

$$\tau \leq latency\_limit$$

Candidates above the limit are rejected before objective comparison.

Solver Modes

Mode	Use case	Behavior
Auto	Default	Enumerates if $\leq$ 12 free nodes; otherwise random search
Enumerate	Small DAGs	Tests every assignment and returns the feasible global best
Random Search	Larger DAGs	Perturbs placements and keeps the best feasible candidate
Simulated Annealing	Escaping local minima	Accepts some worse moves according to temperature

All modes enforce fixed-device nodes and the E2E latency cap.

GUI Workflow

1. Edit nodes, names, compute costs, fixed-device flags.
2. Edit edges and transfer sizes in MB.
3. Configure bandwidth, latency, objective weight, solver mode, iterations, and E2E cap.
4. Click **Solve**.
5. Inspect metrics, topology placement, and profiler-style timeline.

Validation catches duplicate node IDs, unknown edge endpoints, and non-DAG topologies.

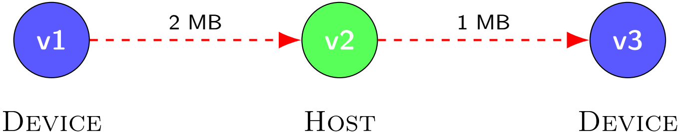
Visualization Design

Topology SVG

- Blue: DEVICE.
- Green: HOST.
- Bold node border: fixed on DEVICE.
- Red dashed edge: cross-device transfer.
- Gray solid edge: local dependency.

Timeline SVG

- Host, Transfer, Device lanes.
- Operator bars use computed start/finish times.
- Transfer bars use network transfer duration.
- Browser-native scroll and scale controls.



The evaluator produces a queue-aware schedule rather than an ideal infinite-parallel critical path.

Current Implementation Stack

- Python 3.9+.
- PyQt6 for desktop controls and layout.
- PyQt6-WebEngine for SVG topology and timeline rendering.
- NetworkX for DAG storage, topological sort, and validation.
- Pytest coverage for evaluator, JSON IO, solver modes, and latency-limit feasibility.

Next Steps

- Add richer solver diagnostics: rejected candidates, constraint slack, convergence history.
- Add exportable SVG/PNG reports for topology and timeline.
- Add batch evaluation over bandwidth/latency sweeps.
- Consider more advanced schedulers for larger DAGs.
- Improve timeline semantics with explicit queue-wait and data-ready intervals.

Questions?